IB Mathematics Application & Interpretation

Higher Level–Prediction Exams

November 2026 - Paper 3(Set-2)

Duration: 1:15 hrs **(Calculator is Allowed)** **Max. Marks: 55**

Question 1

[Maximum mark: 24]

In this question you will analyse probability distributions and perform hypothesis tests with error analysis.

Consider an arithmetic sequence with first term u_1 and common difference d.

It is given that $u_3 = \frac{5}{24}$ and $S_4 = S_6$.

(a) Show that $u_1 = \frac{9}{24}$ and $d = -\frac{1}{12}$. [3]

Answer:

$$s_n = \frac{n}{2}\left[2u_1 + (n-1)\,d\right]$$

$\therefore$ $$S_4 = \frac{4}{2}\left[2u_1 + (4-1)\,d\right] = 2\left(2u_1 + 3d\right)$$

$$S_6 = \frac{6}{2}\left[2u_1 + (6-1)\,d\right] = 3(2u_1 + 5d)$$

$\therefore$ $$2\left(2u_1 + 3d\right) = 3\left(2u_1 + 5d\right)$$

$$4u_1 + 6d = 6u_1 + 15d$$

$$2u_1 + 9d = 0 \quad \text{——— (i)}$$

$$u_3 = u_1 + 2d$$

$$\therefore u_1 + 2d = \frac{5}{24} \quad \text{—— (ii)}$$

Using GDC to solve simultaneous equation (i) & (ii), we have

$$u_1 = \frac{9}{24}, \quad d = -\frac{1}{12}$$

The function f, such that $f(x) = u_1 + (x-1)d$ defines a discrete probability distribution for $1 \leq x \leq n$ where $x, n \in \mathbb{Z}^+$.

(b) (i) Show that $n = 4$. [2]

(ii) Calculate $\sum_{x=1}^{6} f(x)$. [1]

(iii) Explain why $f(x)$ does not define a discrete probability distribution when $n = 6$. [2]

(iv) Complete the following probability distribution table defined by $f(x)$. [1]

x	1	2	3	4
$\mathrm{P}(X = x)$	$\frac{9}{24}$	a	$\frac{5}{24}$	b

Answer:

(b)(i) $f(x) = u_1 + (x-1)d$

$$f(x) = \frac{9}{24} + (x-1)\left(-\frac{1}{12}\right) \quad \text{—— (i)}$$

$$f(1) = \frac{9}{24} + (1-1)\left(-\frac{1}{12}\right) = \frac{9}{24}$$

$$f(2) = \frac{9}{24} + (2-1)\left(-\frac{1}{12}\right) = \frac{9}{24} - \frac{1}{12} = \frac{7}{24}$$

$$f(3) = \frac{9}{24} + (3-1)\left(-\frac{1}{12}\right) = \frac{9}{24} - \frac{2}{12} = \frac{5}{24}$$

$$f(4) = \frac{9}{24} + (4-1)\left(-\frac{1}{12}\right) = \frac{9}{24} - \frac{3}{12} = \frac{3}{24}$$

$$f(1) + f(2) + f(3) + f(4) = \frac{24}{24} = 1$$

Since total probability is 1 for $1 \leq x \leq 4$

$\therefore n = 4$

(ii) $\sum_{x=1}^{6} f(x) = \sum_{x=1}^{6} \left[\frac{9}{24} + (x-1)\left(-\frac{1}{12}\right)\right]$

using GDC

$$\therefore \sum_{x=1}^{6} f(x) = 1$$

(iii) when $n=6$

$f(6) = -\frac{1}{24} < 0$

which is not a probability

(iv)

x	1	2	3	4
$P(X=x)$	$\frac{9}{24}$	$\frac{7}{24}$	$\frac{5}{24}$	$\frac{3}{24}$

Bradley plays a game in which there are four possible outcomes, scoring 1, 2, 3 or 4 points. The outcomes are those found in (b) part (iv), and it is claimed that their associated probabilities are also those found in (b) part (iv).

(c) Explain why a game defined by the terms of an arithmetic sequence, such as this, can never be considered a fair game. [1]

Answer:

$$E(X) = 1\times\frac{9}{24} + 2\times\frac{7}{24} + 3\times\frac{5}{24} + 4\times\frac{3}{24} = \frac{50}{24} = \frac{25}{12}$$

$\therefore E(X) = \frac{25}{12}$

A game is fair if $\sum x P(X=x) = 0$

But $E(X) > 0$

Hence game is not fair

Bradley plays the game 100 times, assuming that the probabilities are correct.

(d) Calculate the probability that Bradley will score 4 points in at least ten games. [3]

Answer: This is a Binomial Distribution

- $n = 100$ (fixed)
- each trial is independent (we assume)
- each trial has two outcomes. Bradley either scores 4 points or does not score 4 points
- Probability of success is constant

$$P(X=4) = \frac{3}{24} = \frac{1}{8} = p \quad \text{(from table)}$$

$\therefore T \sim B(100, \frac{1}{8})$

$\therefore P(T \geq 10) = 0.816$

Bradley believes it is actually much harder to score 4 points than claimed. To test this, Bradley plays the game a further 100 times, to test for the true probability of scoring four points, at the 5% significance level.

His null and alternate hypotheses are

$$H_0 : p = b$$

$$H_1 : p < b$$

(e) (i) Determine the critical region for Bradley's test. [3]

(ii) State the probability that Bradley commits a type I error. [1]

Answer:

$H_0 : p = \frac{1}{8}$

$H_1 : p < \frac{1}{8}$

we require: $P(T \leq t) < 0.05$

Table:

A: t	B: P
0	$= binomcdf\left(100, \frac{1}{8}, 0, A_1\right)$
1	
2	
⋮	
100	= . – – –

t	P(T ≤ t)
0	0.00001587
1	0.00002427
2	0.0001846
3	0.0009332
4	0.003526
5	0.01063
6	0.02672
7	0.05759

Or using graph, we can get the same table:

$$f_1(x) = binomcdf\,(100, 18, 0; int\,(x))$$

And set window settings as follows:

$$x_{\min} = 0, X_{\max} = 100, Y_{\min} = 0, Y_{\max} = 1$$

And create split table, which will give you same probabilities

Note that all the probabilities from t=0 to t=6 are less than 0.05
Hence they are in critical region and we reject null hypothesis.

But $P(T \geq 7) > 0.05$

Therefore it is not within the critical region and we would not reject the null hypothesis in this case.

Hence critical region for Bradley's test is: $T \leq 6$

(i) A type I error occurs when a null hypothesis is rejected when it should not have been. That is , when a value within the critical region is obtained, given that the null hypothesis is true.

i.e. $P(\text{Type I error}) = P(T \text{ within critical region} \mid H_0 \text{ is True})$

In this case, null hypothesis will be rejected when Bradley scores 4 in 6 or less games out of 100. The null hypothesis is p = 1/8.

$$P(\text{Type I error}) = P\left(T \leq 6 \mid p = \tfrac{1}{8}\right)$$

$$\therefore P(\text{Type I error}) = 0.0267$$

Bradley runs the same hypothesis test 10 times.

(f) Calculate the probability that Bradley does not commit a type I error in any of the 10 hypothesis tests. [3]

Answer:

This fits a Binomial Distribution with:

- n =10 (fixed number of trials
- Each trial is independent
- Each trial has two outcomes. Bradley either commits a type I error, or does not commit a type I error
- Probability of success is constant, 0.02672...

Let H represent number of hypothesis tests in which , in which Bradley commits a type I error.

$$\therefore H \sim B\left(10, 0.02672\ldots\right)$$

$$\therefore P\left(H = 0\right) = 0.7626\ldots \approx 0.763$$

[2]

In fact, the true probability of scoring four points is $\mathrm{P}(X = 4) = \frac{1}{20}$.

(g) Calculate the probability that Bradley committed a type II error in his original test. [2]

(h) In the ten hypothesis tests that Bradley runs, calculate the probability that he commits a type II error in the first hypothesis test, but no more after that. [2]

Answer:(g)

Type II error occurs when a null hypothesis is not rejected when it should have been. That is, when a value outside the critical region is obtained, given that the null hypothesis was incorrect.

i.e. $P(\text{Type II error}) = P(T \text{ outside critical region} \mid H_0 \text{ is not true})$

It will happen when Bradley scores a 4 in 7 or more games out of 100. The true probability of scoring 4 points is p=1/20

Hence $T \sim B\left(100, \frac{1}{20}\right)$

$$\therefore P(\text{Type II error}) = P\left(T \geq 7 \mid T \sim B\left(100, \tfrac{1}{20}\right)\right)$$

$$= 0.2339\ldots$$

$$P(\text{Type II error}) = 0.234$$

Answer:(h)

From part (g), we have

$$P(\text{Type II error}) = 0.234$$

Therefore probability of **not** committing a type II error is:

$$P(\text{No type II error}) = 1 - 0.2339\ldots = 0.7660\ldots$$

Because they are complimentary events.

Using the formula for independent events, the probability of Bradley committing a type I error in the first test, but not in the next 9 tests is given by

$$\left(0.2339. \times (0.7660\ldots)^9 = 0.02124. \approx 0.0212\right.$$

Question 2

[Maximum mark: 31]

This question explores the relationship between complex numbers, rotational geometry, and the stability of a predator-prey ecosystem modeled by coupled differential equations.

Consider the complex number $z = x + \mathrm{i}y$, where $x, y \in \mathbb{R}$. A transformation T in the Argand plane maps a point z to a new point w such that $w = (\alpha + \mathrm{i}\beta)z$, where $\alpha, \beta \in \mathbb{R}$.

(a) Given that $w = u + \mathrm{i}v$, show that the transformation T can be written in the matrix form $\boldsymbol{w} = \boldsymbol{Mz}$ as

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \quad [3]$$

Answer:

We have

$$w = (\alpha + i\beta)(x + iy)$$

$$= \alpha x + i\alpha y + i\beta x + i^2 \beta y$$

$$= (\alpha x - \beta y) + i(\beta x + \alpha y) \qquad (\because i^2 = -1)$$

Equating real & imaginary parts

$$u = \alpha x - \beta y$$

$$v = \beta x + \alpha y$$

Writing this as a matrix

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

(b) Let $\lambda = \alpha + \mathrm{i}\beta$.

(i) Show that α can be expressed as $\alpha = |\lambda| \cos(\arg(\lambda))$. [2]

(ii) Find a similar expression for β in terms of $|\lambda|$ and $\arg(\lambda)$. [1]

(iii) Hence, fully describe the geometric effect of the transformation T on the position vector of z, $\begin{pmatrix} x \\ y \end{pmatrix}$. [4]

Answer:

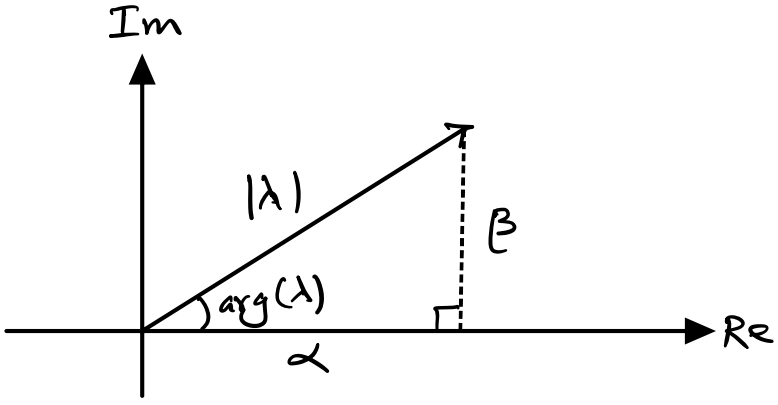
(i) $\alpha = \mathrm{Re}(\lambda)$, $\beta = \mathrm{Im}(\lambda)$

On Argand diagram

In this rt. triangle

$$\cos(\arg(\lambda)) = \frac{\alpha}{|\lambda|}$$

$$\therefore \alpha = |\lambda| \cos(\arg(\lambda))$$



(ii) Similarly,

$$\sin(\arg(\lambda)) = \frac{\beta}{|\lambda|}$$

$$\beta = |\lambda| \sin(\arg(\lambda))$$

(iii) Substituting these values into the matrix of coefficients in place of α and β.

$$M = \begin{pmatrix} |\lambda|\cos(\arg(\lambda)) & -|\lambda|\sin(\arg(\lambda)) \\ |\lambda|\sin(\arg(\lambda)) & |\lambda|\cos(\arg(\lambda)) \end{pmatrix}$$

$$= |\lambda| \begin{pmatrix} \cos(\arg(\lambda)) & -\sin(\arg(\lambda)) \\ \sin(\arg(\lambda)) & \cos(\arg(\lambda)) \end{pmatrix}$$

$$M = \begin{pmatrix} |\lambda| & 0 \\ 0 & |\lambda| \end{pmatrix} \begin{pmatrix} \cos(\arg(\lambda)) & -\sin(\arg(\lambda)) \\ \sin(\arg(\lambda)) & \cos(\arg(\lambda)) \end{pmatrix}$$

Referring to the transformation formula (3.9) we see that the transformation is a combination of an enlargement and an anticlockwise rotation about the origin.

T is an anticlockwise rotation of angle λ about the origin followed by an enlargement of scale factor |λ|.

Marine biologists are studying the interaction between a population of small fish (x) and a species of jellyfish (y) in a closed lagoon. The populations, in thousands, at time t months are modeled by the system of coupled differential equations

$$\frac{dx}{dt} = x - y$$
$$\frac{dy}{dt} = x + y \quad \longrightarrow \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

(c) (i) Write this system in the form $\dot{\boldsymbol{x}} = \boldsymbol{M}\boldsymbol{x}$, where $\boldsymbol{x} = \begin{pmatrix} x \\ y \end{pmatrix}$. [1]

One of the eigenvalues of the matrix $\boldsymbol{M}$ is $\lambda_1 = 1 + \mathrm{i}$.

(ii) Write down the other eigenvalue of the matrix $\boldsymbol{M}$. [1]

(iii) Find the corresponding eigenvectors for these eigenvalues. Give your answers in the form $\boldsymbol{p}_{(1,2)} = \begin{pmatrix} a \\ b \end{pmatrix}$, where $a \in \mathbb{R}^+$ and $b \in \mathbb{C}$. [4]

Answer:

(i) We extract coefficient of x & y to form matrix M

$$\dot{x} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} x$$

(ii) Complex roots always appear in conjugate pairs $a \pm ib$

$$\therefore \quad \lambda_2 = 1 - i$$

(iii) We find eigenvectors for $\lambda_1 = 1 + i$ by solving

$$(M - \lambda I)\vec{P_1} = \vec{0}$$

$$\therefore \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1-\lambda & -1 \\ 1 & 1-\lambda \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1-1-i & -1 \\ 1 & 1-1-i \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} -i & -1 \\ 1 & -i \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

From first row,

$$-ia - b = 0 \Rightarrow -ia = b$$

Since $a > 0$ $\therefore a = 1$ gives $b = -i$

$$\therefore \vec{P_1} = \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

The second eigenvector corresponds to

$\lambda_2 = 1-i$

$$\therefore \begin{pmatrix} 1-(1-i) & -1 \\ 1 & 1-(1-i) \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} i & -1 \\ 1 & i \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

From 1st row, $ai - b = 0 \Rightarrow ai = b$

let $a = 1 \Rightarrow b = i$

$$\therefore \vec{P}_2 = \begin{pmatrix} 1 \\ i \end{pmatrix}$$

(d) Using your results from part (c), show that the general solution for this system is

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = e^t \left[C_1 \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} + C_2 \mathrm{i} \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix} \right]$$

where C_1 and C_2 are real constants. [6]

Answer:

Using exact solution for Coupled Differential Equations formula (5.17), we have

$$x = Ae^{\lambda_1 t}\overrightarrow{p}_1 + Be^{\lambda_2 t}\overrightarrow{p}_2$$

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = Ae^{(1+i)t} \begin{pmatrix} 1 \\ -i \end{pmatrix} + Be^{(1-i)t} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

Taking out common factor

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = Ae^t \cdot e^{it} \begin{pmatrix} 1 \\ -i \end{pmatrix} + Be^t \cdot e^{-it} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

$$= e^t \left[Ae^{it} \begin{pmatrix} 1 \\ -i \end{pmatrix} + Be^{-it} \begin{pmatrix} 1 \\ i \end{pmatrix} \right]$$

Converting it to polar form, we have

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = e^t \left(A(\cos t + i \sin t) \begin{pmatrix} 1 \\ -i \end{pmatrix} + B(\cos(-t) + i \sin(-t)) \begin{pmatrix} 1 \\ i \end{pmatrix} \right)$$

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = e^t \left(\begin{pmatrix} A(\cos t + i \sin t) \\ -iA(\cos t + i \sin t) \end{pmatrix} + \begin{pmatrix} B(\cos t - i \sin t) \\ iB(\cos t - i \sin t) \end{pmatrix} \right)$$

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = e^t \begin{pmatrix} A\cos t + iA\sin t + B\cos t - iB\sin t \\ -iA\cos t - i^2 A\sin t + iB\cos t - i^2 B\sin t \end{pmatrix}$$

$$= e^t \begin{pmatrix} A\cos t + B\cos t + i(A\sin t - B\sin t) \\ A\sin t + B\sin t + i(-A\cos t + B\cos t) \end{pmatrix}$$

$$= e^t \left(\begin{pmatrix} (A+B)\cos t \\ (A+B)\sin t \end{pmatrix} + i \begin{pmatrix} (A-B)\sin t \\ (B-A)\cos t \end{pmatrix} \right)$$

$$= e^t \left((A+B) \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} + (B-A)\, i \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix} \right)$$

Taking $C_1 = A+B$ & $C_2 = (B-A)$, we have

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = e^t \left[c_1 \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} + c_2 i \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix} \right]$$

(e) At time $t = 0$, the population of fish is 1000, and the population of jellyfish is 0.

(i) Find the values of C_1 and C_2. [2]

(ii) Hence, find the population of jellyfish when $t = \frac{\pi}{2}$. [2]

Answer:

Put $t=0$, $x=1$, $y=0$

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} = e^0 \left[c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 i \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right]$$

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} c_1 \\ c_{2i} \end{pmatrix}$$

$\therefore c_1 = 1$, $c_2 = 0$

(ii) For $c_1 = 1$, $c_2 = 0$, general solution is

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = e^t \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}$$

We re only interested in jellyfish component, which is

$$y(t) = e^t \sin t$$

substituting $t = \frac{\pi}{2}$, we have

$$y\left(\frac{\pi}{2}\right) = e^{\pi/2} \sin\left(\frac{\pi}{2}\right)$$

$$= e^{\pi/2}$$

$$= 4.810\ldots$$

Jellyfish population at $t = \frac{\pi}{2}$ is 4810

(f) Describe the long-term behavior of the populations predicted by this model. Relate your answer to the eigenvalues found in (c) part (ii), and the description of the geometric transformations found in (b) part (iii). [3]

Answer:

our exact solution is

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = e^t \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}$$

The vector $\begin{pmatrix} \cos t \\ \sin t \end{pmatrix}$ A circular motion in anticlockwise direction while the factor

e^t Causes population to grow exponentially (spiral outwards/unstable) leading to explosion in population number.

This is consistent with (c) part (ii) where we found complex eigenvalues with positive real parts.
This is indicative of an outward spiral.

It is also consistent with our findings from (b) part (iii) where we found that we had an anticlockwise rotation, with an enlargement by scale factor $|\lambda|$, leading to a spiral.

A critique of the model suggests that the ecosystem will collapse because the populations grow indefinitely. To stabilize the system, the biologists propose introducing a limiting factor such that the eigenvalues of the new system matrix $\boldsymbol{N}$ become purely imaginary (i.e., of the form $\pm ki$).

(g) State the geometric shape of the trajectory of the populations on the phase portrait for this new stable model. [2]

Answer:

If the eigenvalues are purely imaginary, the real part is zero.
The trajectory will be a closed curve / Ellipse / Circle.
This represents stable periodic cycle.

Best of Luck for Exam